

An affordable system for the teleoperation of dexterous robotic hands using Leap Motion hand tracking and vibrotactile feedback

Claudio Coppola¹, Gökhan Solak¹ and Lorenzo Jamone¹

Abstract—Using robot manipulators in contexts where it is undesirable or impractical for humans to physically intervene is crucial for several applications, from manufacturing to extreme environments. However, robots require a high degree of intelligence to operate in those environments, especially if they are not fully structured. Teleoperation compensates for this limitation by connecting the human operator to the robot using human-robot interfaces. The remotely operated sessions can also be used as demonstrations to program more powerful autonomous agents. In this article, we report a thorough user study to characterise the effect of simple vibrotactile feedback on the performance and cognitive load of the human user in performing teleoperated grasping and manipulation tasks. The experiments are performed using a portable and affordable bilateral teleoperation system that we designed, composed of a Leap Motion sensor and a custom-designed vibrotactile haptic glove to operate a 4-fingered robot hand equipped with 3-axis force sensors on the fingertips; the software packages we developed are open-source and publicly available. Our results show that vibrotactile feedback improves teleoperation and reduces cognitive load, especially for complex in-hand manipulation tasks.

I. INTRODUCTION

Bilateral teleoperation plays a crucial role in robotics and immersive Virtual Reality and is fundamental to perform robotic tasks in remote medical, industrial and hazardous environments [1], [2]. The ability to operate robots in extreme environments is fundamental where humans cannot operate physically, such as underwater infrastructure inspection and maintenance, space exploration, or nuclear decommissioning. However, in these contexts, robots are not sufficiently autonomous as the objects and their shapes are not known a priori, and it is impossible to create analytical grasping and manipulation strategies. Furthermore, the robot should handle the objects safely, avoiding any damage or collision [3]. Teleoperation allows humans to intervene directly in these scenarios and perform the task safely [4]. Additionally, remotely operated robot sessions can be used as demonstrations to program more powerful autonomous agents [5] in applications requiring a high level of dexterity (e.g., light manufacturing) [6]. Utilising haptic devices and sensors enables the creation of advanced human-machine interfaces for safe and precise operation. Most available solutions including haptic feedback are complex, expensive,

*Work partially supported by the EPSRC UK through projects NCNR (EP/R02572X/1) and MAN³ (EP/S00453X/1).

¹Claudio Coppola, Gökhan Solak and Lorenzo Jamone are with ARQ (Advanced Robotics at Queen Mary), School of Electronic Engineering and Computer Science, Queen Mary University of London, London, E14NS, UK (emails: {c.coppola, g.solak, l.jamone}@qmul.ac.uk).

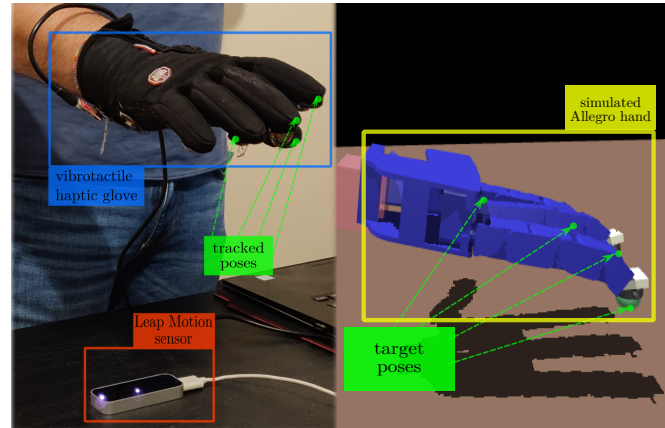


Fig. 1: An overview of the proposed teleoperation approach. On the left, the position of user's fingertips are tracked. The vibrating motors sewed on the glove provide vibrotactile feedback. On the right, the target poses are calculated to control the robot fingertips, equipped with force sensors.

and/or bulky, limiting their potential use in these environments [7]. However, vibrotactile feedback has been used to improve task performance and cognitive load as haptic modality [8]. Basic inexpensive vibrotactile feedback has been shown to be useful in completing simple interactions with virtual objects (e.g. tapping with the finger) in scenarios like playing a flight game [9] or piloting a drone [10]. Similarly, other forms of simple haptic feedback have been used to operate robots for peg in the hole task [11], but without characterising the contribution of their vibrotactile devices (e.g. [11], [12]) to performance and/or cognitive load. Also, user studies are reported to evaluate the experience of the user in recognising tactile patterns as mental imagery or as representation of facial expressions [13], [14].

Hereby, we propose a portable (easy to transport and deploy) and affordable (£200) leader-follower teleoperation system for a dexterous Allegro robot hand equipped with Optoforce 3-axis force sensors on fingertips. On the leader side, we track the human fingers using a Leap Motion sensor (Figure 2b), whose accuracy has been extensively tested [15]. We map the movements to the robot in the cartesian space, with no need for a user-specific calibration procedure. A light wearable glove (Figure 2c) provides haptic feedback using Eccentric Rotating Mass (ERM) motors mounted on its fingertips [16], [17]. On the follower side, the Allegro Hand has Optoforce 3-axis force sensors on the fingertips. The force feedback is sent back to the user as vibrotactile stimulation.

Our solution has been qualitatively tested with the real robot¹, achieving successful grasping and in-hand manipulation of objects. However, for a quantitative system evaluation, we conducted a user study with a realistic physics simulation of the robot (using the Mujoco² software), and analysed the benefits of the vibrotactile feedback, with multiple naive users. This is the first study to empirically evaluate the impact of simple vibrotactile haptic feedback on performance and cognitive load while performing grasping and manipulation tasks in this setting. The main contributions of the paper are:

- 1) a thorough user study that indicates the following:
 - (I) vibrotactile feedback improves teleoperation performance during challenging grasping tasks in high precision settings.
 - (II) vibrotactile feedback improves teleoperation performance during dexterous in-hand manipulation tasks, by increasing the effective workspace, and decreasing the time needed to complete the tasks.
 - (III) vibrotactile feedback reduces the cognitive load of the user in teleoperation of grasping and manipulation tasks, especially in terms of *Effort* and *Physical Demand*, and it increases the perceived *Performance* as measured using the NASA TLX metrics.
- 2) a new open-source framework for the teleoperation³ of dexterous multi-fingered hands based on Leap Motion hand tracking and vibrotactile feedback⁴;

The paper is organised as follows: in Section II, we present the state of the art for teleoperation of manipulators based on hand tracking. In Section III, we introduce our teleoperation framework for a four-fingered robot hand and illustrate the details of the control system. In Section IV, we describe the experimentation setting and the tasks used to evaluate our approach. In Section V, we present the results of the experiments in detail. In Section VI, we conclude by discussing our approach and results and presenting possible directions for future research in this area.

II. RELATED WORK

A bilateral teleoperation setup is composed of two parts: an input system, to track the movements of the human user and transfer them to the robot; and a feedback system, to convey information about the remote environment back to the user. The literature contains a wide spectrum of solutions and devices to provide a human interface and force feedback [18]. One first aspect to consider is the *type of devices involved* and thus, the type of input and feedback employed.

On one side, we can find solutions based on **exoskeletons** [19] which provide high-fidelity position tracking and "true" feedback to the user. However, those tend to be bulky and extremely expensive. Other approaches, rely on more affordable **sensorised gloves** [20]. These are less expensive than the previous, but they can still achieve high-quality

input and very reliable force feedback modalities, which are simulated with high-quality actuators. The authors in [21], show a bilateral teleoperation system for a soft hand based on air compression piloted with a data glove. [22] present a teleoperation system for a soft gripper using a sensing glove equipped with soft haptic actuators. In [23], the authors propose a Cybertouch data glove-based teleoperation of a 3D-printed hand. [24] presents a glove-based solution in which the hand state is tracked with 15 accelerometer units. Other recent approaches focused on inexpensive haptic devices such as vibration motors or displays. These have been successfully employed in creating tactile experiences and interfaces [25], [13], [14], or in teleoperation applications [11], [9], [12]. However, No user study has effectively shown the effect of inexpensive haptic devices on either cognitive load or success rate while performing object grasping/manipulation tasks with dexterous hands.

Recent approaches use hand tracking techniques based on **Computer Vision** [26] using either depth sensors [27], [28], RGB cameras [29] or arrays of RGB-D cameras [28]. Those compromise on mapping precision with lower infrastructural requirements and lower cost. To provide force feedback, these approaches rely on pseudo-haptic feedback [30] or additional dedicated hardware [31], [32].

Teleoperation systems are also characterised by the type of *robot manipulator*, their finger quantity and the choice of the *mapping algorithm*. Those are strongly connected as more complex manipulators (i.e. higher DoF) require sophisticated mapping schemes. In the most basic case, a pre-designed **direct mapping** is created between the joints of the human and the robot. For Example, [27] directly maps the joints tracked with a Leap Motion sensor to control a 2 fingered gripper, while [23] uses the synergies computed on the first 2 principal components of the joints tracked on a Cyber-Touch glove. Solutions based on **kinematic retargeting** map between the joint spaces of the user and the robot using constrained optimisation. For example, [28] searches the joint configuration of a dexterous robot, minimising the difference in fingertip-distances between the user's and robot's hand using least-squares optimisation.

The mapping can be performed on the Cartesian space using **Jacobian matrix inversion**, e.g., to operate a Pisa/IIT SoftHand on a UR5 arm [32]. **inverse kinematic solvers** can be used to calculate the joint states from the Cartesian pose of the end-effector avoiding the numerical issues of Jacobian inversion, e.g., to map the position of the user's thumb and index to a 3-fingered hand robot [33] or to map for a DLR-HIT II hand in combination with an RBF-NN [34].

To abstract from the kinematic differences of leader and follower systems a object-based mapping can be created by learning a control synergy between the operator and robot hand. This is particularly useful when kinematic chains are asymmetric, although it may reduce the generalisation of movements [35], [11], [36].

In our approach, we are teleoperating the fingertips of a 4-fingered dexterous manipulator with a very affordable device set. The user's hand is tracked with a Leap Motion

¹Video available: <https://youtu.be/xiJxB5OeEs8>

²<http://www.mujoco.org/>

³Repo Available https://github.com/ARQ-CRISP/teleoperation_ur5_allegro_leap

⁴Repo Available https://github.com/ARQ-CRISP/haptic_glove_ros

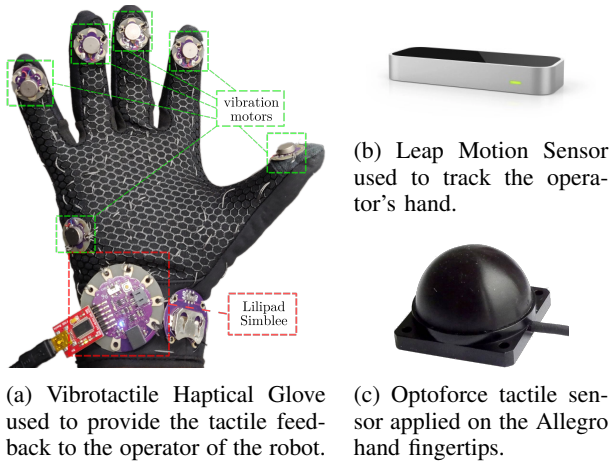


Fig. 2: The devices used in this work

sensor, and the force feedback is provided as a vibration on the fingertips by a custom data glove. Our main objective is to demonstrate that also affordable technologies can permit the effective teleoperation of a dexterous robot. In particular, the user study shows that the simple vibration feedback we provide is better than no feedback at all.

III. METHOD

Our robot setting consists of an Allegro hand mounted on a UR5 6-DOF arm (Figure 3). The UR5 arm is attached to a custom-built metal infrastructure to keep its workspace at the right height. The Allegro hand has 4 fingers with 4 independently controllable joints each. The fingertips are equipped with Optoforce OMD 20-SE-40N sensors (Figure 2C) capable of measuring 3-dimensional force with 1 kHz frequency. The fingertip target positions are followed using a PID position controller. We also compensate the gravity effects on the robot hand. More details can be found in [37].

The teleoperation hardware consists in the Leap Motion (Figure 2B) depth sensor used to track the position of the fingertips of the operator and a customly designed haptic glove (Figure 2A) providing vibrotactile feedback when the Optoforce sensor detects some pressure.

A. Affordable Haptic Data Glove

The user is given tactile feedback using the data glove shown in Figure 2A. The data glove is made by sewing 6 vibration motors and a LilyPad Simblee board on a textile glove. The board connects the motors to the system via ROS serial⁵. All components are tied to each other using conductive threads. The board is connected to the PC using an FTDI header that provides both the power and the serial communication. The ROS-compatible driver code for this device have been made publicly available⁴. While the motors can vibrate in any frequency between 0 and 255 Hz, in this study, only the 0 and 255 levels of vibrations are used, as the study is performed on a simulated environment and the

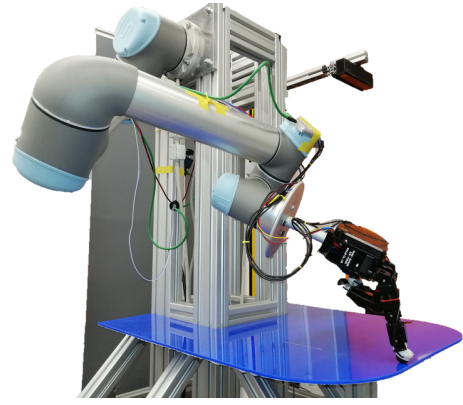


Fig. 3: Photo of the real MARIA robot platform. The platform is composed of an Allegro hand mounted UR5 robot arm attached to a customly designed metal infrastructure.

simulated tactile sensor is less sensitive than the real one. Tactile feedback events of the motors are triggered when the magnitude of the force sensor readings are above 7.5 mN. These motors are very low-cost, limiting the resolution and vibration frequencies supported by the device and its responsiveness. These limits make the perception of haptic feedback problematic. For this reason, the feedback is delivered in a binary fashion. In this study, only the vibrators corresponding to the first 4 fingers (*thumb, index, middle, ring*) are used in accordance with the sensors available on the robot hand. Additional information about the glove can be found in [17].

B. Cartesian Space Teleoperation

The proposed bilateral teleoperation system maps the kinematics of the fingers of the operator onto the fingers of the teleoperated robot. We implemented this mapping approach on ROS and made it publicly available³.

For every finger $f \in \{\text{thumb, index, middle, ring}\}$, both the position R^f of the robot fingertips and the tracked positions of the operator's fingertips H^f are represented in the frame of reference of their wrists. From now on, we will assume that those two frames overlap so that robot and human fingers can be represented in the same frame of reference. The mapping algorithm generates the target position and orientation of each fingertip. The first is generated with a combination of position and velocity of the tracked fingertips. However, as the Leap Motion sensor does not provide the finger orientation explicitly, it is estimated from the direction of the distal phalanges (i.e. the bones at the tips of the fingers) of the operator.

The position of the robot fingertips R^f are generated combining the position and velocity of the fingertips H^f with a weighted average of the targets obtained with each approach:

$$\hat{R}^f = \beta \times \hat{R}_{pos}^f + (1 - \beta) * \hat{R}_{vel}^f \quad \text{with } \beta \in [0, 1] \quad (1)$$

In this formulation, $\hat{R}^f$ corresponds to the combined position target for the allegro fingertip, $\hat{R}_{pos}^f$ is the target position generated through position-based mapping, $\hat{R}_{vel}^f$ is the target

⁵<http://wiki.ros.org/roserial>

position generated through velocity-based mapping and β is a mixing factor used to decide the importance of the two components. In this study, the position is considered more reliable, thus β is set to 0.95.

The position-based mapping generates the targets $\hat{R}_{pos}^f$ adapting the current position of the operator's fingertips H^f by scaling it to the robot hand's size. Thus, when a hand is detected, it estimates the length of the finger L^f by summing the length of each of its sections L_i^f provided by the Leap Motion tracker. Knowing the length of the finger of the robot A^f , the targets are generated as follows:

$$\hat{R}_{pos}^f(t) = H^f(t) \times \frac{A^f}{L^f} \gamma^f \quad (2)$$

where γ^f is used to adjust the value of the ratio and it can be different for each finger. In this study, we chose $\gamma^{thumb} = 1$ and $\gamma^f = 1.6$ for the other fingers after a process of trial and error. The γ^f coefficient exists to compensate ulterior structural differences in size that do not depend directly on the fingers. For example, $\gamma^f = 1.6$ in the primary fingers, to account for the size of the Allegro Hand palm.

The velocity-based mapping, instead, uses the operator's fingertip movements ΔH^f to update the position of the fingertip of the robot hand. Once a new hand is detected by the teleoperation software the position of the robot hand $R^f(t_0)$ is sampled for every finger $f \in \{thumb, index, middle, ring\}$ and set as the initial target position $\hat{R}^f(t_0)$. When a new sample of the tracked hand arrives, ΔH^f is computed and consequently added to the current position of each finger R^f . The new target position is computed as following:

$$\begin{aligned} \Delta H^f &= H^f(t + \Delta t) - H^f(t) \\ \hat{R}^f(t + \Delta t) &= \hat{R}^f(t) + \alpha^f * \Delta H^f \end{aligned} \quad (3)$$

where α^f is a scaling factor used to take into account the impact that the difference in the size of the robot and human hand has on the fingertip velocity. The downside of velocity mapping is that the coherence of the fingertip mapping depends on the positions $H^f(t_0)$ and $R^f(t_0)$, with t_0 being the time when the user's hand is detected. Therefore, if not paired with a calibration protocol, the target positions may diverge. This problem is solved by using it together with the position mapping. Indeed, the target $\hat{R}^f(t_0)$ is computed using the position mapping only, creating a good initial position for the velocity mapping.

Since the Leap Motion does not provide the orientations of the fingertips, we calculate the target orientations Q_a^f using the tracked joint positions corresponding to the extremes of the finger distal phalanges H_{distal}^f and H_{tip}^f . Where H_{distal}^f is the position of the distal joint and H_{tip}^f is the position of the fingertip. The fingertip orientation is the orientation of the vector $(H_{tip}^f - H_{distal}^f)$ with respect to the wrist frame. The roll angle is set to a fixed value θ chosen in coherence with the configuration of the robot hand. The value of θ is different for the primary fingers (*index, middle, ring*) and for the *thumb*. In the first case, $\theta = \pi/2$, while for the thumb $\theta = \pi$. Finally, we apply a tilt about the x -axis by $\pi/4$ as the Optoforce sensors are mounted with that angle.

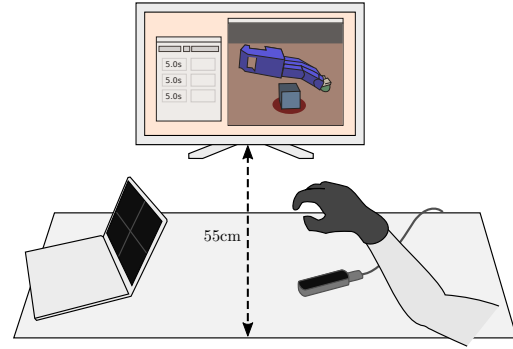


Fig. 4: Experimental setup: the screen displays the robot simulation. The participant places the hand above the Leap Motion sensor while wearing the haptic glove. On the left, the operator monitors the experiment through a laptop.

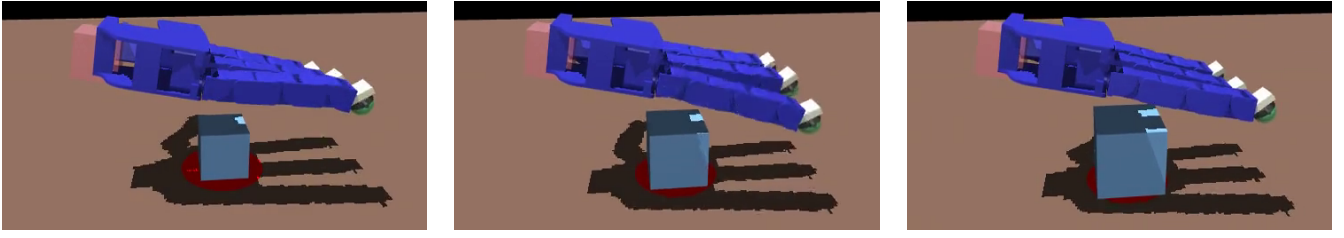
IV. EXPERIMENTS

We performed experiments to test the effect of having vibrotactile feedback while teleoperating robot fingers. In particular, we examine whether the Vibrotactile Haptic feedback reduces the user's physical demand and effort during the teleoperation in specific scenarios like grasping smaller objects and performing in-hand manipulation.

The experiment consists of performing multiple tasks and measuring user performance. The task is repeated 10 times for each scenario, with and without haptic feedback, on 3 differently sized objects. In total, we recorded 60 episodes for each subject. For this experiment, we recruited 10 subjects within the university staff through direct request. At the end of the experiments, the participants are asked to anonymously fill in 4 NASA-TLX questionnaires [38] to measure the task load index. These are used to measure the cognitive load on the **Grasping** and **Manipulation** sub-tasks using only vision (**Vision**) and vision plus haptic feedback (**Haptic**).

The experiment setup is illustrated in Figure 4. The users sit in front of a screen showing the simulated robot hand and the object to grasp. On the side, the experiment operator loads different simulated scenarios and initiates the experiment protocol by the graphical interface visible on the screen of the user. The interface also shows the time remaining for each of the sub-tasks the participant has to perform. The leap motion sensor is placed on a desk situated in between the screen and the participant who sits at 55cm distance from the screen. The participants are asked to wear the haptic glove in both cases (with and without the haptic feedback) as the presence of the glove could impact the performance of the leap motion hand tracking. Furthermore, wearing a glove can limit the mobility of users, especially if the glove size does not fit them. The recording starts after 1 or 2 trials to get the participant acquainted with the teleoperation system.

The teleoperation system has been tested on the MARIA robot on a set of real objects¹, but we implement our experiments on a physics simulator to guarantee consistency of measurements and initial positions. As the participants are



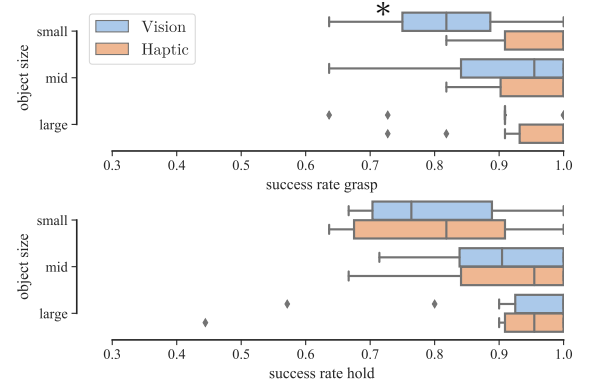
(a) **Small** cube (2.5cm per side) scenario. (b) **Middle** cube (3cm per side) scenario. (c) **Large** cube (3.5cm per side) scenario.

Fig. 5: Experimental setup: size of the different cubes.

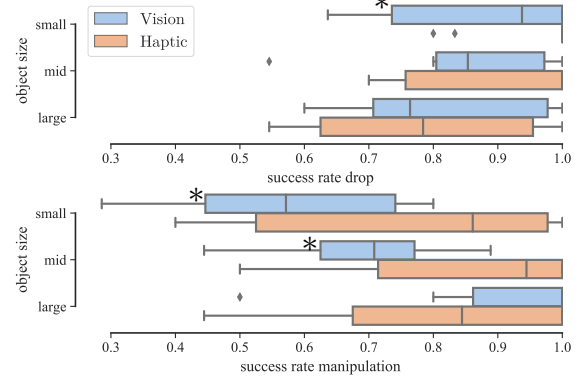
not previously trained, using the simulation prevents damage to the physical robot. We chose Mujoco for our simulations, as it compares favourably to other frameworks [39]. We created a 3D model of the Allegro hand and hand-tuned the model parameters to achieve a realistic behaviour; we also added the objects and a circular red marker appearing under the simulated objects.

The participants are asked to perform a single task divided in three sequential sub-tasks: **Grasp**, **Hold**, **Manipulate**. During **Grasp** the participant is asked to perform a hand closure to grasp the shown object. After 5s, the robot hand automatically lifts the object 15cm high. Afterwards, the participant is asked to **Hold** the object for 5s without letting it fall. At the end of this time interval, the robot hand is lowered by 10cm. Finally, the participant is asked to **Manipulate** the object. During this phase, the participant drops the object as far as possible from the initial position (marked with a red circle of 4.5cm) within 5s. After that time, the task ends. During these tasks, the user has no control over the pose of the hand, but just of the fingers. The task is performed by each participant in 2 modalities: **Vision** and **Haptic**. In the first case, the participant has access only to the visual information of the simulated environment from a static camera. In the second, the participant receives the visual information and haptic feedback from the glove. The objects used during experiment are cubes (Figure 5) of 3 different sizes and the same weight and friction coefficient. The first (**Small**) is a cube of 2.5cm per side, the second (**Middle**) 3cm and the third (**Large**) of 3.5cm.

Finally, to measure the success of the task we define an ordinal metric. Its levels correspond to the success in performing the defined sub-tasks. Because of the nature of the tasks, performing one sub-task successfully implies success in the previous ones (i.e., succeeding in holding the object implies succeeding in grasping it). The performance metric levels are **fail**, **grasp success**, **hold success**, **drop success** and **manipulate success**. The performance metric level semantics are as follows: a) **fail** - a failure in grasping the object within the 5s time limit; b) **grasp success** - a grasp capable of lifting the object without falling; c) **hold success** - the object is held for 5s after the arm is lifted; d) **drop success** the object is dropped within the 5s limit; e) **manipulate success** - the centre of the dropped object is outside of the circular marker in the simulation (4.5cm of radius).



(a) box plot for the success rate of the grasp and hold sub-tasks



(b) box plot for the success rate of the drop and manipulate sub-tasks

Fig. 6: Results of the task success rate analysis, partitioned by object size and modality. ♦ indicates outliers and * indicates statistically relevant difference in performance (see Table II).

V. RESULTS

In this section, the experiment results are analysed to evaluate the effect of the haptic feedback on the performance and the experience of teleoperation. In particular, statistical tests and data analysis are designed to verify that vibrotactile haptic feedback supports grasping when working with smaller objects, requiring more precision (Contribution (I)), that it enhances user's performance in performing dexterous manipulation (Contribution (II)) and that it reduces the cognitive load on the user in terms of Effort and Physical

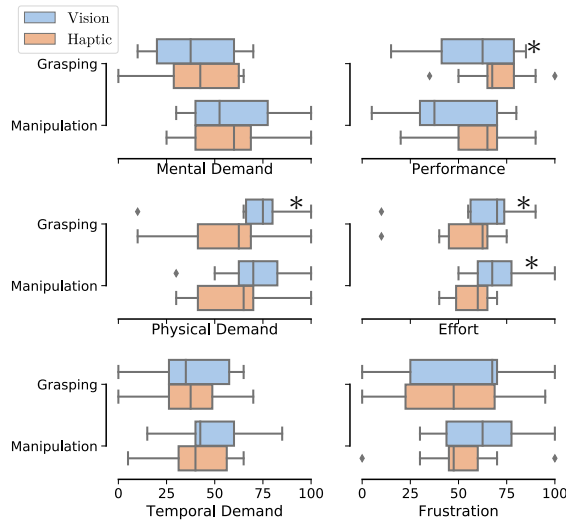


Fig. 7: Box plots of the NASA-TLX Questionnaire results. ♦ indicates outliers and * indicates statistically relevant difference (see Table I). The results are partitioned by task (Grasping and Manipulation) and the sensing modality.

Demand (Contribution (III)). To verify those statements the analysis will be partitioned by modality and object size.

A first analysis is performed considering the success levels described in Section IV. Aggregating the data along with the participants and the trials, we can create a comparison between the trials performed with the 2 different modalities stratified for the object sizes. In Figure 6, a boxplot is used to compare the 2 modalities in the 4 defined levels of success. The success rate is computed ignoring the failures that occurred at the previous stages to avoid biasing the results (e.g trials in which the participant failed grasping the object, are not used for the statistics of hold, drop and manipulation). In Figure 6a, can be observed the statistics of the success rate for the **grasp** and **hold** sub-tasks. The haptic feedback does not have a particular effect on the capability to **hold** the object. However, it results advantageous for performing the grasp in all the cases. Figure 6b shows a performance improvement in **dropping** the small objects within the time limit. Furthermore, in the **manipulation** task, the haptic feedback creates a statistically relevant improvement on the small and middle-sized objects. To examine in which cases the success of the tasks are statistically dependant on the modality we use the χ^2 -test for independence. The p-values and χ -scores of the significant cases are reported in Table II and highlighted in Figure 6 with a * symbol.

Figure 7 shows median and confidence intervals for the metrics of cognitive load defined in NASA-TLX test [38]. According to the graph, it is possible to see that the haptic feedback provides an improvement in terms of *Performance*, *Effort* and *Physical Demand*. The significance of this result is tested with a one-sided Wilcoxon test as it is not limited to Normal distributions. This test is chosen because it is not bound to Gaussian distributions and because it is designed to deal with ordinal measures sampled from the

Metric	Task	Rank score	p-value
Physical Demand	Grasping	36.0	0.006
Performance	Grasping	8.5	0.047
Effort	Grasping	33.0	0.018
Effort	Manipulation	39.0	0.02

TABLE I: Table containing the p-value and the rank value obtained with a single-sided Wilcoxon test. Only the cases in which p-value < 0.05 are reported on the table this table.

Task	Size	p-value	χ^2
grasp success	small	0.004	8.274
drop success	small	0.039	4.228
manipulation success	mid	0.005	7.681
manipulation success	small	0.009	6.692

TABLE II: Table containing the p-value and the χ^2 value obtained with a single-sided χ^2 -test for independence. Only the cases in which $p < 0.05$ are reported on this table.

same population. As the *Performance* component of the TLX is the only "higher-better" metric of the TLX, it is tested on the other tail of the distribution. The results in Table I, shows that the improvement is statistically significant in all the 3 aforementioned components for the **grasping** and only the *Effort* component during the **manipulation**. Therefore, the haptic feedback increases the confidence of the operator in positioning the fingertips, reducing the effort spent to obtain the right position for the grasp. Similarly, during the manipulation, the haptic feedback lowers the effort to re-position the object with just the fingertips.

It has to be noted, that there is high variability and many outliers in the presented results. This is partly the consequence of using low-cost components. Indeed, many mapping failures depend on the noise of the hand tracker, increased by the presence of the glove.

VI. CONCLUSIONS

This article reports a user study which shows that a basic vibrotactile haptic feedback improves the quality of dexterous robotic teleoperation. In particular, it indicates that the haptic feedback significantly improves the grasping performance on objects requiring additional precision (e.g. smaller); that it enhances the performance during dexterous in-hand manipulation tasks; that vibrotactile feedback significantly reduces the *Effort* requested from the user while operating the hand to perform either grasping or manipulation tasks. During grasping tasks, also *Physical Demand*, and perceived *Performance* also improve significantly. Future studies will replicate these experiments on the real robot and will use the developed system for learning by demonstration tasks. The experience collected during this work will also be used to improve the teleoperation device set. In particular, wearing a glove, especially of dark colour, reduces the performance of the hand tracker. Additionally, an unfit glove size limits the user's capability to teleoperate. For future iterations, we will consider using alternative supports for the vibration motors, such as thimbles.

REFERENCES

- [1] P. F. Hokayem and M. W. Spong, "Bilateral teleoperation: An historical survey," *Automatica*, pp. 2035–2057, 2006.
- [2] M. Wonsick, P. Long, A. Ö. Önel, M. Wang, and T. Padir, "A holistic approach to human-supervised humanoid robot operations in extreme environments," *Frontiers in Robotics and AI*, vol. 8, p. 148, 2021.
- [3] R. Smith, E. Cucco, and C. Fairbairn, "Robotic development for the nuclear environment: Challenges and strategy," *Robotics*, vol. 9, no. 4, 2020. [Online]. Available: <https://www.mdpi.com/2218-6581/9/4/94>
- [4] O. Tokatli, P. Das, R. Nath, L. Pangione, A. Altobelli, G. Burroughes, E. T. Jonasson, M. F. Turner, and R. Skilton, "Robot-assisted glovebox teleoperation for nuclear industry," *Robotics*, vol. 10, no. 3, p. 85, 2021.
- [5] W. Si, N. Wang, and C. Yang, "A review on manipulation skill acquisition through teleoperation-based learning from demonstration," *Cognitive Computation and Systems*, vol. 3, no. 1, pp. 1–16, 2021.
- [6] P. Tsarouchi, S. Makris, and G. Chryssoulouris, "Human–robot interaction review and challenges on task planning and programming," *International Journal of Computer Integrated Manufacturing*, pp. 916–931, 2016.
- [7] C.-J. Yang, J.-F. Zhang, Y. Chen, Y.-M. Dong, and Y. Zhang, "A review of exoskeleton-type systems and their key technologies," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, pp. 1599–1612, 2008.
- [8] M. J. Massimino and T. B. Sheridan, "Sensory substitution for force feedback in teleoperation," in *Analysis, Design and Evaluation of Man–Machine Systems 1992*. Elsevier, 1993, pp. 109–114.
- [9] F. Sanfilippo and C. Pacchierotti, "A low-cost multi-modal auditory-visual-tactile framework for remote touch," in *2020 3rd International Conference on Information and Computer Technologies (ICICT)*. IEEE, 2020, pp. 213–218.
- [10] J. Footitt, D. Brown, S. Marks, and A. M. Connor, "Development of a wearable haptic game interface," *EAI Endorsed Trans. Creative Technol.*, vol. 3, p. e5, 2016.
- [11] L. Meli, G. Salvietti, G. Gioioso, M. Malvezzi, and D. Prattichizzo, "Multi-contact bilateral telemanipulation using wearable haptics," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2016, pp. 1431–1436.
- [12] P. Galambos and P. Baranyi, "Vibrotactile force feedback for telemanipulation: Concept and applications," in *2011 2nd International Conference on Cognitive Infocommunications (CogInfoCom)*. IEEE, 2011, pp. 1–6.
- [13] Y. Muramatsu, M. Niitsuma, and T. Thomessen, "Perception of tactile sensation using vibrotactile glove interface," in *2012 IEEE 3rd International Conference on Cognitive Infocommunications (CogInfoCom)*. IEEE, 2012, pp. 621–626.
- [14] S. Krishna, S. Bala, T. McDaniel, S. McGuire, and S. Panchanathan, "Vibroglove: an assistive technology aid for conveying facial expressions," in *CHI'10 Extended Abstracts on Human Factors in Computing Systems*, 2010, pp. 3637–3642.
- [15] F. Weichert, D. Bachmann, B. Rudak, and D. Fisseler, "Analysis of the accuracy and robustness of the leap motion controller," *Sensors*, pp. 6380–6393, 2013.
- [16] B. Junput, X. Wei, and L. Jamone, "Feel It on Your Fingers: Dataglove with Vibrotactile Feedback for Virtual Reality and Telerobotics," in *Towards Autonomous Robotic Systems*, K. Althoefer, J. Konstantinova, and K. Zhang, Eds. Cham: Springer International Publishing, 2019, pp. 375–385.
- [17] B. Junput, I. Farkhatdinov, and L. Jamone, "Touch it, rub it, feel it! haptic rendering of physical textures with a low cost wearable system," in *Towards Autonomous Robotic Systems (TAROS)*, 2020.
- [18] R. Li, H. Wang, and Z. Liu, "Survey on mapping human hand motion to robotic hands for teleoperation," *IEEE Transactions on Circuits and Systems for Video Technology*, 2021.
- [19] M. Sarac, M. Solazzi, and A. Frisoli, "Design requirements of generic hand exoskeletons and survey of hand exoskeletons for rehabilitation, assistive, or haptic use," *IEEE transactions on haptics*, vol. 12, no. 4, pp. 400–413, 2019.
- [20] D. Wang, M. Song, A. Naqash, Y. Zheng, W. Xu, and Y. Zhang, "Toward whole-hand kinesthetic feedback: A survey of force feedback gloves," *IEEE Transactions on Haptics*, pp. 189–204, 2019.
- [21] M. Li, Y. Zhuo, J. Chen, B. He, G. Xu, J. Xie, X. Zhao, and W. Yao, "Design and performance characterization of a soft robot hand with fingertip haptic feedback for teleoperation," *Advanced Robotics*, pp. 1491–1505, 2020.
- [22] J. H. Low, W. W. Lee, P. M. Khin, N. V. Thakor, S. L. Kukreja, H. L. Ren, and C. H. Yeow, "Hybrid tele-manipulation system using a sensorized 3-d-printed soft robotic gripper and a soft fabric-based haptic glove," *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 880–887, 2017.
- [23] Q. Zhu, V. P. da Fonseca, B. M. R. Lima, M. Welyhorsky, M. Goubran, T. E. A. de Oliveira, and E. M. Petriu, "Teleoperated grasping using a robotic hand and a haptic-feedback data glove," in *2020 IEEE International Systems Conference (SysCon)*. IEEE, 2020, pp. 1–7.
- [24] B. Fang, F. Sun, H. Liu, and D. Guo, "A novel data glove using inertial and magnetic sensors for motion capture and robotic arm-hand teleoperation," *Industrial Robot: An International Journal*, 2017.
- [25] D. Prattichizzo, F. Chinello, C. Pacchierotti, and K. Minamizawa, "Remotouch: A system for remote touch experience," in *19th International Symposium in Robot and Human Interactive Communication*. IEEE, 2010, pp. 676–679.
- [26] H. Cheng, L. Yang, and Z. Liu, "Survey on 3d hand gesture recognition," *IEEE transactions on circuits and systems for video technology*, pp. 1659–1673, 2015.
- [27] V. P. da Fonseca, B. M. R. Lima, T. E. A. de Oliveira, Q. Zhu, V. Z. Groza, and E. M. Petriu, "In-hand telemanipulation using a robotic hand and biology-inspired haptic sensing," in *2019 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*. IEEE, 2019, pp. 1–6.
- [28] A. Handa, K. Van Wyk, W. Yang, J. Liang, Y.-W. Chao, Q. Wan, S. Birchfield, N. Ratliff, and D. Fox, "Dexpilot: Vision based teleoperation of dexterous robotic hand-arm system," *arXiv preprint arXiv:1910.03135*, 2019.
- [29] F. Gomez-Donoso, S. Orts-Escolano, and M. Cazorla, "Accurate and efficient 3d hand pose regression for robot hand teleoperation using a monocular rgb camera," *Expert Systems with Applications*, vol. 136, pp. 327–337, 2019.
- [30] M. Achibet, B. Le Gouis, M. Marchal, P.-A. Leziart, F. Argelaguet, A. Girard, A. Lécuyer, and H. Kajimoto, "Flexifingers: Multi-finger interaction in vr combining passive haptics and pseudo-haptics," in *2017 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, 2017, pp. 103–106.
- [31] Y. Fan, C. Yang, and X. Wu, "Improved teleoperation of an industrial robot arm system using leap motion and myo armband," in *2019 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. IEEE, 2019, pp. 1670–1675.
- [32] J. Bimbo, C. Pacchierotti, M. Aggravi, N. Tsarakakis, and D. Prattichizzo, "Teleoperation in cluttered environments using wearable haptic feedback," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2017, pp. 3401–3408.
- [33] I. Jang, J. Carrasco, A. Weightman, and B. Lennox, "Intuitive bare-hand teleoperation of a robotic manipulator using virtual reality and leap motion," in *Annual Conference Towards Autonomous Robotic Systems*. Springer, 2019, pp. 283–294.
- [34] C. Li, A. Fahmy, and J. Sienz, "Development of a neural network-based control system for the dlr-hit ii robot hand using leap motion," *IEEE Access*, pp. 136914–136923, 2019.
- [35] G. Salvietti, L. Meli, G. Gioioso, M. Malvezzi, and D. Prattichizzo, "Multicontact bilateral telemanipulation with kinematic asymmetries," *IEEE/ASME Transactions on Mechatronics*, vol. 22, no. 1, pp. 445–456, 2016.
- [36] G. Gioioso, G. Salvietti, M. Malvezzi, and D. Prattichizzo, "An object-based approach to map human hand synergies onto robotic hands with dissimilar kinematics," in *Robotics: Science and Systems VIII*. The MIT Press Sydney, NSW, 2012, pp. 97–104.
- [37] G. Solak and L. Jamone, "Learning by demonstration and robust control of dexterous in-hand robotic manipulation skills," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 8246–8251.
- [38] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research," in *Human Mental Workload*, ser. Advances in Psychology, P. A. Hancock and N. Meshkati, Eds. North-Holland, 1988, pp. 139–183.
- [39] T. Erez, Y. Tassa, and E. Todorov, "Simulation tools for model-based robotics: Comparison of bullet, havok, mujoco, ode and physx," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 4397–4404.